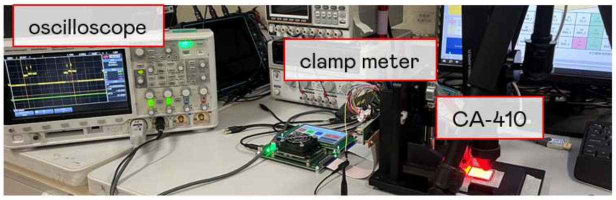


Figure 5. Measurement setup.

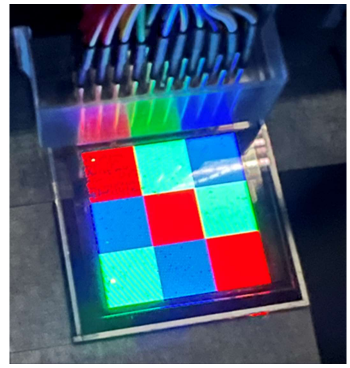
3. Results and Discussion

As shown in Figure 6, experimental results demonstrate that the proposed CST driving scheme successfully maintains the optical performance while significantly reducing the required driving current. Under the same driving conditions, the luminance and chromaticity produced by CST are nearly identical to those obtained using the standard RGB driving method. For example, the measured point under standard RGB operation was approximately 629 nits with a chromaticity of (0.196, 0.071), whereas CST, when operating only in the first sub-frame, achieved 628 nits with a chromaticity of (0.192, 0.070). The deviation in both luminance and color coordinates remained minimal, indicating that time-division operation of CST does not compromise the perceived image quality.

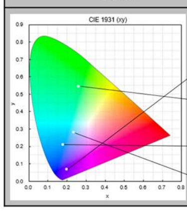
Beyond maintaining optical characteristics, CST provides a substantial reduction in panel current. When driving above-mentioned point (0.196, 0.071), the conventional RGB approach required about 62.5 mA, whereas CST reduced this value to approximately 35.0 mA using only the first sub-frame, corresponding to a current reduction of about 44% and power reduction of about 25%. Similar improvements were observed for other color point, for example, grass green (0.258, 0.547), where the current decreased from roughly 33.8 mA to 20.0 mA. For other composite color such as sky blue (0.177, 0.212), CST operating with both sub-frames reduced the total current from the standard 52.5 mA to 40.1mA, representing about 24% in current saving and 10% in power saving. For composite color close to white, (0.226, 0.284), standard RGB driving consumes 332mA while CST only consumes 150mA, corresponding to 55% of current saving and 33% of power saving. Although the degree of improvement varies with the color composition, the trend consistently shows that CST lowers energy consumption effectively.

The underlying mechanism for these reductions can be attributed to several factors inherent to CST. First, by dividing a single frame into two sub-frames, CST avoids the simultaneous activation of all sub-pixels, thereby reducing the instantaneous load on the power lines. Second, by exploiting current recombination - such as using parts of the green and blue currents to compensate the red emission - CST produces equivalent luminance with a lower effective current.

Despite the structural differences in its driving method, CST does not introduce visible artifacts such as luminance instability, or chromaticity drift. Since both sub-frames operate at high frequency, users perceive the output as a single stable frame with uniform brightness. The close alignment of optical results between CST and standard RGB indicates that the human visual system effectively integrates both sub-frames, confirming that CST preserves image quality while reducing power consumption.



(a)



Color	Standard RGB		Color-Sequential Tandem			
	Lum. & CIE (x, y)	Total current	Lum. & CIE (x, y)	Total current	Current saving	Power saving
	629 nits (0.196, 0.071)	62.5 mA	628 nits (0.192, 0.070)	35.0 mA	-44%	-25%
	1310 nits (0.258, 0.547)	33.8 mA	1328 nits (0.258, 0.550)	20.0 mA	-41%	-21%
	1552 nits (0.177, 0.212)	52.5 mA	1528 nits (0.178, 0.211)	40.1 mA	-24%	-10%
	5835 nits (0.226, 0.284)	332 mA	5229 nits (0.231, 0.285)	150 mA	-55%	-33%

(b)

Figure 6. (a) CST test sample light up. (b) Current and power measurement of standard RGB driving vs. CST driving under different color combination.

4. Conclusion

This study introduces a Color Sequential Tandem (CST) driving method for micro-LED panels. The scheme divides a single frame into two sub-frames, utilizing different emission paths in each to effectively reduce instantaneous current while preserving luminance and chromaticity, without incurring additional LED costs. Experimental results demonstrate a power reduction of approximately 10% to 33% across various colors.

CST provides a low-power driving strategy for micro-LED displays, particularly in applications requiring ultra-high brightness and low energy consumption such as HUDs. Future work may extend CST evaluation to higher-resolution panels and more complex display content such as PHUD and 3D AR-HUDs to further validate its applicability in automotive products.

5. References

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